

EQUATIONS IN AND OVER GROUPS

(E1) Describe the groups over which any equation is solvable.

(E2) (M.Kervaire, F.Laudenbach) Let $F_n/R = \langle x_1, \dots, x_n \mid r_1, \dots, r_m \rangle$ be a presentation of a non-trivial group. Is it true that a group $\langle x_1, \dots, x_n, x_{n+1} \mid r_1, \dots, r_m, s \rangle$ is also non-trivial for any element s from F_{n+1} ? [Background](#)

*(E3) (L.Comerford) If an equation over a free group F has no solutions in F , is there a finite quotient of F in which the equation has no solutions? (If so, this provides another proof of Makanin's theorem). [Background](#)

(E4) (J.Birman) Let $F = F_n$ be the free group of rank n generated by $a_1, \dots, a_n$. Is there a solution of the equation $y_1 a_1 y_1^{-1} \dots y_n a_n y_n^{-1} = a_1 \dots a_n$ with all y_i from the second commutator subgroup F'' ? (The answer is "no" if and only if the Gassner representation of the pure braid group P_n is faithful -- cf. problem [\(B2\)](#)).

(E5) (G.Baumslag, A.Miasnikov, V.Remeslennikov) Is a free product of two equationally noetherian groups equationally noetherian? (A group is called equationally noetherian if every system of equations in this group is equivalent to a finite subsystem). [Background](#)

(E6) (G.Baumslag, A.Miasnikov, V.Remeslennikov) Is a free pro- p group equationally noetherian?